LESSON 6.10

Divide Mixed Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing with mixed numbers means, rewrite them as improper fractions, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Chen is mapping a suspect's route. The total distance is 3 1/2 miles, and each segment between landmarks is 1/4 mile. She needs to know how many segments are on the route to place surveillance cameras at every landmark.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mixed Number *Spanish: Número mixto*	A whole number plus a fraction, like 2 1/3.	2 1/3 means 2 wholes and 1/3 more — picture 2 full circles and 1/3 of another
Improper Fraction *Spanish: Fracción impropia*	A fraction where the top is bigger than or equal to the bottom, like 7/4.	7/3 is improper because 7 > 3. It equals 2 1/3 as a mixed number.
Convert *Spanish: Convertir*	To change a number to a new form but keep the same value.	2 1/3 → multiply 2 × 3 = 6, add 1 → 7/3. Same value, different form.
Simplify *Spanish: Simplificar*	To write a fraction with smaller numbers that names the same amount, like 2/4 = 1/2.	8/6: GCF is 2 → 8÷2 / 6÷2 = 4/3 = 1 1/3
Reciprocal *Spanish: Recíproco*	A fraction turned upside down. Dividing by a fraction gives the same answer as multiplying by its reciprocal.	The reciprocal of 3/4 is 4/3. To divide by 3/4, multiply by 4/3.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Agent Chen has a 3 1/2-mile route cut into 1/4-mile segments. I want 3 1/2 ÷ 1/4.

1. First I convert 3 1/2. Multiply the whole by the bottom: 3 × 2 = 6. Add the top: 6 + 1 = 7. So 3 1/2 = 7/2.
2. Now I ask how many 1/4s fit in 7/2. In fourths, 7/2 = 14/4, and 14 fourths ÷ 1 fourth = 14 — the common-denominator method.
3. The reciprocal shortcut gives the same answer: 7/2 × 4/1 = 28/2 = 14.
4. Check by multiplying: 14 × 1/4 = 14/4 = 3 1/2. So 3 1/2 ÷ 1/4 = 14 segments.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's try 2 1/2 ÷ 1/2. First convert 2 1/2: 2 × 2 = 4, plus 1 = 5, so 2 1/2 = 5/2.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Before dividing, I convert 2 1/2 to _____ because it is easier to _____ with improper fractions.

Word bank Mixed Number | Improper Fraction | Convert | Simplify | Reciprocal | improper | 5/2 | convert | multiply | reciprocal | The reciprocal shortcut

WATCH OUT!

A common mistake

A common mistake in Divide Mixed Numbers is dividing the whole-number parts and fraction parts separately instead of converting first

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Divide Mixed Numbers

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Agent Chen is mapping a suspect's route. The total distance is 3 1/2 miles, and each segment between landmarks is 1/4 mile. She needs to know how many segments are on the route to place surveillance cameras at every landmark.

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Convert each mixed number to an improper fraction.

Mixed Number	Improper Fraction
1 3/4	
2 2/3	
4 1/2	
3 1/5	

HINT Every conversion keeps its denominator — 1 3/4 stays in fourths.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 1 1/2 ÷ 3/4?

(A) 1 1/8
(B) 2
(C) 3/4
(D) 9/8

HINT 1 1/2 is a mixed number — it has to become one improper fraction before you divide.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 2 1/4 ÷ 3/4?

(A) 1 1/2
(B) 2 1/3
(C) 3
(D) 9/16

HINT Convert 2 1/4 to an improper fraction before anything else.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is 3 1/2 ÷ 1/4?

(A) 3 1/8
(B) 7/8
(C) 13/4
(D) 14

HINT Two steps: convert 3 1/2 first, then divide by 1/4.

SHOW YOUR WORK

PART 3 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

Convert 3 2/5 to an improper fraction.

(A) 6/5

(B) 15/5

(C) 17/5

(D) 32/5

HINT Converting keeps the denominator 5 the same — only the top number changes.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

PART 4 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** 1 1/2 ÷ 3/4
2. **Convert** 1 1/2 = 3/2
3. **Multiply by the reciprocal** 3/2 × 3/4
4. **Multiply** 9/8 = 1 1/8

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at the reciprocal step — did the divisor get flipped?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why is converting to an improper fraction the first step when dividing mixed numbers?

SENTENCE STARTER Before dividing, I convert 2 1/2 to ___ because it is easier to ___ with improper fractions.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.10

Divide Mixed Numbers

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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WHY THIS MATTERS Agent Chen is mapping a suspect's route. The total distance is 3 1/2 miles, and each segment between landmarks is 1/4 mile. She needs to know how many segments are on the route to place surveillance cameras at every landmark.

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Convert each mixed number to an improper fraction, then divide.

Problem	Convert to Improper	The reciprocal shortcut	Quotient
3 1/2 ÷ 1/4			
1 3/4 ÷ 1/2			
2 1/3 ÷ 7/6			
4 1/5 ÷ 3/5			

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each mixed number division to its quotient.

_____ 2 1/4 ÷ 3/4
_____ 4 1/2 ÷ 9/8
_____ 1 2/3 ÷ 5/6
_____ 3 1/3 ÷ 2/3
_____ 5 1/4 ÷ 3/4
_____ 1 1/2 ÷ 1/4

(A) 3
(B) 4
(C) 2
(D) 5
(E) 7
(F) 6

3 **Use the model to solve. Show your work.**



How many 1/2-size pieces fit into 2 1/2?

SHOW YOUR WORK

Answer: _____

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A baker has 2 1/4 cups of butter. Each batch of cookies needs 3/4 cup of butter. How many batches can the baker make?

(A) 1 1/2 batches
(B) 2 batches
(C) 3 batches
(D) 6 batches

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Agent Chen has walked 2 1/4 miles, which is 3/4 of her whole route. How long is the whole route? (2 1/4 ÷ 3/4 asks how much is in one whole route.)

(A) 1 11/16 miles
(B) 3 miles
(C) 1/3 mile
(D) 2 1/3 miles

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** 3 1/3 ÷ 2/3
2. **Multiply by the reciprocal (skipped converting)** 3 1/3 × 3/2
3. **Multiply the fraction part only** 1/3 × 3/2 = 3/6 = 1/2
4. **Combine with the whole** 3 + 1/2 = 3 1/2

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why is converting to an improper fraction the first step when dividing mixed numbers?

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LESSON 6.10

Divide Mixed Numbers

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS Agent Chen is mapping a suspect's route. The total distance is 3 1/2 miles, and each segment between landmarks is 1/4 mile. She needs to know how many segments are on the route to place surveillance cameras at every landmark.

1 **Find the mistake. Explain it, then show the correct work.**

Find the Mixed Number Error

1. **Problem** 2 1/3 ÷ 1/2
2. **Convert** 2 1/3 = 6/3
3. **Multiply by the reciprocal** 6/3 × 2/1
4. **Multiply** 12/3 = 4

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

2 **Answer in complete sentences.**

A detective has a rope that is 5 1/4 feet long. She needs to cut it into pieces that are each 3/4 foot for tying evidence tags. How many pieces can she cut? Write the equation, show each step (convert, the reciprocal shortcut, multiply, simplify), and verify your answer.

WRITE YOUR ANSWER

3 Find the mistake. Explain it, then show the correct work.

Find Liam's Mistake

1. **Problem** 3 1/2 ÷ 1 1/4
2. **Convert first number** 3 1/2 = 7/2
3. **Convert second number** 1 1/4 = 5/4
4. **Divide** 7/2 ÷ 5/4 = 7/2 × 5/4 = 35/8 = 4 3/8

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why is converting to an improper fraction the first step when dividing mixed numbers?

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